AIDED

Assisted Analysis of the Dynamic Electron Density

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1 Introduction

This is a test introduction section.

2 Calculation of the static Electron Density

The (static) Electron Density (ED) of a molecule with M positively charged nuclei and N negatively charged electrons gives the probability of finding an electron in the infinitesimally small space $d\mathbf{r_x}$ at the point $\mathbf{r_x} = (x, y, z)$:

$$\rho(r_x) = N \int \Psi(r_x, \mathbf{r}_2, \ldots, \mathbf{r}_N; \mathbf{R}) \Psi^*(r_x, \mathbf{r}_2, \ldots, \mathbf{r}_N; \mathbf{R}) \, d\mathbf{r}_2 \ldots d\mathbf{r}_N$$

where Ψ is the spinless ground-state 'wave function' for the N electrons, distributed in the field of nuclei with fixed positions (given by the 3M vectors $\mathbf{R}$).

Given an equlibrium geometry $\mathbf{R}$ and accompanying ground-state wave function Ψ, a common practice is to describe the many electron wavefunction, Ψ, as an antisymmetric linear combination of one-electron Molecular Orbitals (MOs), Φ_i, each of which is described by a linear combination of Atomic Orbitals (AO), ϕ_j.

Each AO is expanded in Gaussian Type Orbitals (GTO), g_k, centered at a nucleus such that

$$g_k(x, y, z) = (x - X_k)^{n_k} (y - Y_k)^{m_k} (z - Z_k)^{l_k} \, \mathrm{Exp} \left[-\alpha_k \left| \mathbf{r} - \mathbf{R}_k \right|^2 \right]$$

giving rise to the expressions for AOs and MOs as:

$$\phi_j(\mathbf{r}) = \sum_k d_{jk} g_k(\mathbf{r})$$

$$\Psi_i(\mathbf{r}) = \sum_j c_{ij} \phi_j(\mathbf{r})$$

The final ED can then be expressed as:

$$\rho(\mathbf{r}) = \sum_{\mu,\nu} \sum_{\mu,\nu} P_{\mu\nu} \phi_\mu(\mathbf{r}) \phi_\nu(\mathbf{r})$$

where $P_{\mu\nu}$ is the density matrix, given by $P_{\mu\nu} = 2\sum_k c_{\mu k} c_{\nu k}$ for closed system shells.

TODO: Go into more detail on this calculation.

3 MSDA: Mean Square Displacement Amplitudes

The normal mode analysis of a system containing N vibrating nuclei in thermal equilibrium leads to a 3N-multivariate normal distributino of nuclear displacements relative to their equilibrium positions ($\mathbf{u} = \mathbf{R} - \mathbf{R}^0$):

$$P(\mathbf{u}) = (2\pi)^{-3N/2} |\mathbf{U}|^{-1/2} \exp\left(-\frac{1}{2}\mathbf{u}\mathbf{U}^{-1}\mathbf{u}^T\right)$$

where $\mathbf{u}^T$ represents the transpose of $\mathbf{u}$ and $\mathbf{U}$ is the covariance matrix of the displacements.

This covariance matrix is the Mean Square Displacement Amplitude (MSDA) matrix associated with the expectation values of the Cartesian nuclear displacement products (second moments):

$$\mathbf{U} = \langle \mathbf{u}\mathbf{u}^T \rangle_T$$

The temperature dependence of $\mathbf{U}$ is embedded in the MSDAs of the normal modes in its eigenvalues $\delta = (\delta_{i=1,3N})$:

$$\mathbf{U} = \mathbf{L}\delta\mathbf{L}^T$$

$$\delta_j = \frac{h}{8\pi^2 \nu_j} \coth\left(\frac{h\nu_j}{2k_B T}\right)$$

where ν_j is the frequency of the j-th normal mode, h is Planck's constant, and k_B is Boltzmann's constant.